SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID –** 26041
- **Problem Statement Title-** AR-Based Vocational Training Simulator for Industrial Safety in Jharkhand's Mining & Manufacturing Sector
- **Theme-** Smart Education (Vocational & Skill Training)
- **Sub-Theme-** Blockchain & Cybersecurity (Tamper-Proof Verification)
- **PS Category-** Software
- **Team ID-** 53476
- **Team Name (Registered on portal) :** SHADOW STACK



SHADOW STACK

IDEA TITLE: SafeAR

AI-Powered Semi-Offline AR Safety Training & Verifiable Certification

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❖ Proposed Solution (Describe your Idea/Solution/Prototype)

• Detailed explanation of the proposed solution :

➤ **Android WebAR Simulator:** Zero-headset 3D interactive training platform running directly in mobile browsers on standard ₹10k smartphones.

➤ **Semi-Offline Edge Architecture:** Preloaded site drill packages run 100% offline in deep underground mine shafts without internet.

➤ **Local Cryptographic .dat DB:** NIST SHA-256 Merkle-sealed snapshots ensure training data integrity even in completely disconnected zones.

➤ **Blockchain Micro-Credentials:** Automated on-chain certificate minting with public QR verification (/verify) against fraud.

• How it addresses the problem :

➤ **Replaces Static Textbooks:** Transforms passive 100-page paper manuals into hands-on spatial muscle-memory drills for hazard response.

➤ **Solves Tribal Language Divide:** Full vernacular voice & HUD translation in Santali (`sat`) and Hindi (`hi`) for Jharkhand's workforce.

➤ **Eliminates Internet Requirement:** Underground pits have zero Wi-Fi; SafeAR completes training and scoring 100% offline at the edge.

➤ **Competency Over Attendance:** AI state-engine grades real decision-making, sequence adherence & reaction times, stopping fake attendance.

• Innovation and uniqueness of the solution :

STEP 1: HAZARD IDENTIFICATION — Detect methane vents, loose rock strata & electrical hazards via interactive 3D WebGL.

STEP 2: MANDATORY PPE VALIDATION — Worker selects correct hard hat with headlamp & high-vis vest before entering danger zone.

STEP 3: 5-STEP LOTO ISOLATION — Turn off disconnect switch, lock out, tag out, dissipate residual energy & test zero energy.

STEP 4: AI SCORING & BLOCKCHAIN MINT — Instant on-device score calculation, SHA-256 digest sealing & Ethereum credential issuance.

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TECHNICAL APPROACH

Edge-First Architecture: Train Offline in Deep Mines, Sync Securely to Blockchain

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• Technologies to be used (e.g. programming languages, frameworks, hardware) :

➤ **Frontend & 3D Simulation Engine :** React 18, TypeScript, Three.js (WebGL 60 FPS hardware acceleration), Tailwind CSS, Vite, Lucide Icons.

➤ **Security & Cryptography :** NIST FIPS 180-4 SHA-256 Hashing, Merkle Root Integrity Trees, PBKDF2/Salt Key Derivation, Tamper-Evident Digest.

➤ **Blockchain & Decentralized Ledger :** Ethereum / EVM Smart Contracts (Solidity), Ethers.js, ERC-721/Verifiable Credential Schema, Public QR Lookup.

➤ **Offline Persistence & Cloud Edge :** Encrypted .dat Binary Snapshots, LocalStorage Edge DB, Cloud Firestore (Offline Fallback Cache), Vercel Edge CDN.

• Methodology and process for implementation (Flow Charts/Images/ working prototype) :

MODE 1: SEMI-OFFLINE (Deep Mine)

1. Colliery Arrival: Pit head, 0% Wi-Fi.
2. Preloaded Drills: Local 3D assets.
3. WebGL Drill: Hazard & PPE practice.
4. On-Device AI: Evaluates sequence & time.
5. SHA-256 Digest: Sealed in .dat DB.
6. Surface Sync: Queued for upload.

MODE 2: ONLINE (Classroom / ITI)

1. Induction Portal: Worker / Trainer login.
2. Vernacular Toggle: Santali, Hindi, Tamil.
3. 3D AR Simulator: Active Audio HUD.
4. Instant AI Grade: 0-100% competency.
5. Ethereum Ledger: Direct minting.
6. QR Certificate: Tamper-proof PDF.

MODE 3: FIELD AUDIT (DGMS Official)

1. Spot Inspection: Worker badge check.
2. QR Code Scan: Zero-login lookup.
3. Public Verifier: Open /verify/:id.
4. On-Chain Check: Matches tx hash.
5. Merkle Root Match: Genuine pass.
6. Tamper Rejection: Forgeries flagged.

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FEASIBILITY AND VIABILITY

Engineered for Resource-Constrained Workplaces & Scalable Multi-Site Deployment

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• Analysis of the feasibility of the idea :

➤ **Technical Feasibility:** Proven WebGL Three.js stack running smoothly at 60 FPS on mid-range Android 10+ devices; zero ₹50k VR headset dependency.

➤ **Operational Feasibility:** Preloaded asset bundles execute 100% offline in deep underground shafts; launches in under 10 seconds.

➤ **Economic Feasibility:** Uses workers' existing personal smartphones; 75% cheaper than physical mine training mock chamber setups.

➤ **Software Testing Proof:** 100 Passed / 100 Automated Unit Tests across all 12 test suites (Auth, Cryptography, Three.js, Blockchain).

• Commercial Viability & Policy Alignment :

➤ **Market Opportunity:** 5.5+ Lakh mining and heavy manufacturing workforce in Jharkhand & Eastern India; ₹3,500 Cr industrial safety training market.

➤ **Statutory Alignment:** Directly addresses NEP 2020 Vocational Education (TVET), DGMS Coal Mines Regulations 2017, and Skill India safety mandates.

➤ **Investment & ROI:** Minimal deployment cost; projected break-even within 12 months; 20x ROI via reduction in mine accident compensation and lost-time injuries.

➤ **Multi-Site Scalability:** Modular JSON scenario engine expands seamlessly to metal smelting, chemical handling, and fire rescue.

• Potential challenges and risks & Strategies for overcoming these challenges :

1. Underground Low Light & Tracking Failure

• Potential Risk: Dark coal chambers and featureless walls disrupt optical AR camera tracking. • **Strategy: Dual-Mode Architecture: Seamless fallback to high-contrast WebGL virtual simulation mode with touch orbit controls requiring zero camera tracking.**

2. Worker Literacy & Technology Hesitation

• Potential Risk: Tribal contract workers in Jharkhand struggle with written English instructions. • **Strategy: Multilingual Voice HUD: Native Santali and Hindi audio instructions + intuitive 1-tap color-coded visual touch buttons.**

3. Offline Data Tampering in Deep Mines

• Potential Risk: Malicious actors could tamper with local offline scores before synchronizing. • **Strategy: NIST SHA-256 Merkle Sealing: Any single-byte modification in .dat file triggers instant cryptographic tamper rejection during surface sync.**

IMPACT AND BENEFITS

Quantifiable Life-Saving Outcomes, Economic Value & Regional Empowerment

• Potential impact on the target audience :

➤ **Mining & Manufacturing Workers:** Rehearsing emergency drills builds instinctive spatial muscle memory, reducing workplace fatalities by an estimated 60% to 70%.

➤ **Contract & Tribal Labourers:** Empowers non-English speaking workers through native Santali and Hindi training, promoting equitable safety education.

➤ **Safety Trainers & Supervisors:** Automated competency analytics replace manual paperwork, freeing up 15+ hours per week per trainer.

➤ **Government Inspectors (DGMS):** Instant, zero-login QR code verification at pit heads eliminates counterfeit safety certificates.

• Benefits of the solution (social, economic, environmental) :

➤ **Social & Inclusion Gain:** Bridges the digital divide for Jharkhand's tribal mining workforce; fosters certified, employable professionals in line with NEP 2020.

➤ **Economic & Financial Gain:** Saves ₹100s of crores in mine shutdown penalties, heavy equipment breakdown, and accident compensation; 75% cheaper than mock drills.

➤ **Environmental & Safety Gain:** Reduces hazardous methane leaks and fire explosions through proactive procedural mastery, safeguarding mining ecosystems.

• Feature Comparison Matrix: SafeAR vs Traditional Industrial Training Methods

Evaluation Metric	Paper Manuals	Bulky VR Rigs	SafeAR (Our Solution)
Hardware Requirement	None (Static Paper)	₹50k+ Expensive Headset	✅ **Any ₹10k Android Phone**
100% Offline Underground Operation	✅ Yes (Ineffective)	❌ No (Needs PC/Wi-Fi)	✅ **Yes (.dat Edge DB)**
Vernacular Santali & Hindi Support	❌ English Only	❌ English Only	✅ **Yes (Santali & Hindi)**
Tamper-Evident Verification	❌ Paper (Easily Forged)	❌ Local App Records	✅ **NIST SHA-256 + Ethereum**
Cost per Trained Worker	Low (Zero Muscle Memory)	Very High (Prohibitive)	✅ **Minimal (Software Scalable)**

RESEARCH AND REFERENCES

Source Basis, Technical Standards & Live Evaluator Production Demos

- **Details / Links of the reference and research work :**

1. Statutory Policy Framework & Regulatory Basis :

➤ **DGMS Coal Mines Regulations 2017 & Vocational Training Rules 1966:** Mandates structured, verifiable safety drills prior to entering underground mine pits. Link: https://dgms.gov.in

➤ **NEP 2020 & Skill India Mission:** Mandates technical vocational education (TVET) integration, experiential learning-by-doing, and vernacular inclusivity. Link: https://www.education.gov.in/nep

➤ **NIST FIPS 180-4 Secure Hash Standard:** Provides mathematical basis for SHA-256 cryptographic password hashing and tamper-evident Merkle roots. Link: https://csrc.nist.gov/publications/detail/fips/180-4/final

➤ **IEEE & OSHA Occupational Safety Research:** Studies prove interactive 3D simulations increase emergency hazard retention from 20% to over 75%. Link: https://ieeexplore.ieee.org

2. Verified Production Demos, Repositories & Working Prototype Links :

- **Live Production Application :** https://sihprototype-ten.vercel.app

(Three-Role Authentication: Worker, Safety Trainer, Enterprise Admin)

- **Interactive 3D Simulation :** https://sihprototype-ten.vercel.app/training/mining-basic-scene

(60 FPS WebGL Mining Excavator, LOTO Switch, Emergency Gas Drill & Atmospheric Dust)

- **Public Blockchain Credential Verifier :** https://sihprototype-ten.vercel.app/verify/SAFEAR-2026-DEMO01

(Zero-login public consensus verification with on-chain Ethereum transaction hash)

- **Automated Software Test Suite :** 100/100 Passed Unit Tests across 12 test suites verifying offline persistence, cryptographic hashes, and Three.js engine.
- **Official Organization Sponsor :** Department of Higher & Technical Education, Government of Jharkhand — Problem Statement 26041